Banking and programming of fluency

Two or three questions in 5 to 8 minutes, correction included.
the prerequisites are available; the table offers spaced repetitions, to be adjusted according to the errors.
The ritual does not replace problems or research time.

Issues by first use

R01 · Calculation and indices. First sequences; accessible from S2. For $u_n = 3n - 2$, calculate u_4 then u_{4+1}.

R02 · Thoughtful calculation. Powers; accessible from S2. Calculate 2^{-3} and $\frac{2^5}{2^3}$ No decimals.

R03 · Context. Proportions; accessible from S2. A decrease of 20%, then increases by 20 per cent. Is it finding its original value?

R04 · Vectors. Chasles; accessible from S3. Simplify $\overrightarrow{AB} + \overrightarrow{BC} - \overrightarrow{AC}$.

R05 · Contact details. Subtraction; accessible from S3. For A=(1;2), B=(4;-1), calculate $\overrightarrow{AB}$ and midpoint.

R06 · Colinarity. Proportionality; accessible from S3. Compare vectors $(2;3)$ and $(-4;-6)$.

R07 · Python. For buckle; accessible from S4. What values k takes in range(2.5) ?

R08 · Indices. Amounts; accessible from S4. How many terms contains $\sum_{k=3}^{8} u_k$?

R09 · Thoughtful calculation. Geometric sequence; accessible from S4. Calculate $1 + 2 + 4 + 8 + 16$. Provide Two methods.

A10 · Logic. Quantifiers; accessible from S5. Denying "all terms are positive or null."

A11 · Evidence. Recurrence; accessible from S5. What stage is missing if we have only proved $P(n) \Rightarrow P(n+1)$?

R12 · Logic. Involvement; accessible from S5. The converse of "x=2 implies $x^2 = 4$" is it True?

R13 · Vectors. Combination; accessible from S6. Develop 2 $(\vec{\imath} + \vec{k}) - 3(\vec{\jmath} + \vec{k})$.

R14 · Geometry. Base; accessible from S6. Can a triplet of coplanar vectors be a space base?

R15 · Error. Perspective; accessible from S6. Two lines cut on the drawing of a solid. Are the straight lines necessarily stinging?

R16 · Inequalities. Order; accessible from S7. From $1 \leq x < 4$, deduce a framework from $-2x + 3$.

A17 · Evidence. Recurrence; accessible from S7. In $\sum_{k=0}^{n+1} k$, what term is added to the sum up to n?

R18 · Counter-example. Monotonicity; accessible from S7. To give an increasing follow-up, including all terms are negative.

R19 · Method. Affine sequence; accessible from S11. For $u_{n+1} = u_n/2 + 3$, what a constant subtract to get a geometric sequence?

R20 · Calculation. Fractions; accessible from S11. Simplify $\frac{1}{n+1} - \frac{1}{n+2}$ for $n \geq 0$.

R21 · Geometry. Plan; accessible from S12. Are two directions sufficient to determine A plan?

R22 · Reading tape. Absolute value; accessible from S14. Translate $|u_n - 3| < 0{,}1$ double inequality.

R23 · Threshold. Inequation; accessible from S14. What is the first natural integer such as $1/(n+1) < 1/10$?

R24 · Method. Limits; accessible from S16. For $(3n^2+1)/(n^2+2n)$, what transformation Choose?

R25 · Limit. Geometric sequence; accessible from S19. Compare limits of 0.8 ^{n} and $(-0{,}8)^n$.

R26 · Counter-example. Convergence; accessible from S20. Does a bounded sequence converge Always?

R27 · Theorem. Monotonous convergence; accessible from S20. A increasing sequence increased by 5 Does it necessarily have a limit of 5?

R28 · Calculation. Roots; accessible from S21. Simplify $\sqrt{x^2}$ when x<0.

A29 · Sign. Quotient; accessible from S23 What is the sign of 1 $/(x-2)$ where $x < 2$? limit in 2 $^-$?

R30 · Method. Conjugation; accessible from S24. Transform $\sqrt{x+1} - 1$ to study the quotient by x in 0.

R31 · Choice. Counting; accessible from S22. Three highs and two pants: how many outfits?

R32 · Order. Lists; accessible from S25. How many three-letter codes between A,B,C with Repeat?

R33 · No repetition. Lists; accessible from S25. Same question without repetition.

R34 · Method. Comparative growth; accessible from S26. Give the limit of x^2e^{-x} in $+\infty$.

R35 · Derivation. Product; accessible from S29. Dirty $(x+1)e^x$.

A36 · Error. Composition; accessible from S29. Correct the proposed derivative $((2x+1)^3)' = 3(2x+1)^2$.

R37 · Graph. Tangente; accessible from S29. A tangent passes through (0;1) and (2;5). Its slope?

R38 · Reading. Convexity; accessible from S29. If $f'' < 0$ on I, what can we say about f?

R39 · Counter-example. Inflection; accessible from S29. $f''(0) =$ Is 0 enough for inflection?

R40 · Continuity. Point value; accessible from S31. For $g(x) = (x^2 - 1)/(x - 1)$ out of 1, What value in 1 for continuity?

R41 · Theorem. TVI; accessible from S32. What adds strict monotonicity to the given existence By the VIT?

R42 · Python. Dichotomy; accessible from S32. After three divisions by two of an interval of width 2, what is its width?

R43 · Combinatory. Parts; accessible from S30. Count teams of two out of five.

R44 · Identity. Pascal; accessible from S35. Calculate $\binom{5}{2} + \binom{5}{3}$ without factorial.

R45 · Logic. Additional; accessible from S35. Why count "all but none" for "at least one"?

R46 · Probabilities. Conditioning; accessible from S36. If P(A)=0.4 and $P_A(B) = 0,3$, calculate $P(A \cap B)$.

R47 · Probabilities. Inversion; accessible from S36. Numbers $P_A(B)$ and $P_B(A)$ Are they still equals?

R48 · Event. Integer; accessible from S36. Give the opposite of "at least three successes".

R49 · Model. Binomial; accessible from S38. List the conditions of a Bernoulli scheme.

R50 · Probabilities. Binomial; accessible from S38. For X of law $\mathcal{B}(5; p)$, give P(X=0).

R51 · Average. Binomial; accessible from S39. For X of law $\mathcal{B}(20; 0{,}3)$, give E(X) and V(X).

R52 · Bornes. Probabilities; accessible from S41. Express P(X>4) with a cumulative probability.

R53 · Bornes. Probabilities; accessible from S41. Express $P(2 \leq X \leq 5)$ by difference in cumulations.

R54 · Domain. Logarithm; accessible from S44. Determine the domain of ln $(3 - x)$.

R55 · Calculation. Logarithm; accessible from S44. Simplify ln 12 $-$ ln 3.

R56 · Error. Logarithm; accessible from S44. Test ln $(a + b) = \ln a + \ln b$ with a=b=1.

R57 · Derivative. Logarithm; accessible from S47. Diririr ln $(1 + x^2)$.

R58 · Limit. Logarithm; accessible from S49. Give the limit of $x \ln x$ in 0^+.

R59 · Inequality. Threshold; accessible from S50. In $q^n < c$ with 0<q<1 and 0<c<1, what meaning after division by ln q?

R60 · Contact details. Scalar product; accessible from S48. Calculate $(1;2;0)\cdot(3;-1;4)$ in coordinates Orthonormous.

R61 · Angle. Scalar product; accessible from S48. A negative scalar product indicates what type between two non-zero vectors?

R62 · Normal. Plan; accessible from S51. How many non-colonial directions of the plan do we need? Test against a normal candidate?

R63 · Projection. Geometry; accessible from S54. State the two conditions of one foot of projection on a plan.

R64 · Inverse calculation. Primitives; accessible from S52. Give a antiderivative of 3 e^{2x}.

R65 · Error. Primitives; accessible from S53. Can two antiderivatives over one interval differ from a non-constant function?

R66 · Model. ED; accessible from S55. Resolve y·=2y with y(0)=3.

R67 · Units. Context; accessible from S55. Convert 1.5 h to minutes, then 0.1 min to seconds.

R68 · Python. Euler; accessible from S55. For y-y, y(0)=1 and h=0.1, calculate the first step d-Euler.

R69 · Sign. Complete; accessible from S59. A negative function on [a;b], a<b, can it have a positive integral?

R70 · Calculation. Comprehensive; accessible from S61. Calculate $\int_0^1 2x\,\mathrm{d}x$.

R71 · Units. Comprehensive; accessible from S63. Integrate flow in L/min for minutes Which unit?

R72 · Medium. Comprehensive; accessible from S63. If $\int_2^5 f = 12$, what is the average on [2;5]?

R73 · Transformation. Variance; accessible from S62. If V(X)=5, give V(2X+3).

R74 · Medium. Independence; accessible from S64. For n independent copies of variance 9, give the variance of the average.

R75 · Error. Dispersion; accessible from S64. Multiply the size of an independent sample By 4 divides its standard deviation by how much?

R76 · Parameter. Right; accessible from S65. For M(t)=(1+t;2-t;3t), give one point and one point Director.

R77 · Normal. Plan; accessible from S67. Give a normal plan 2x-y+3z=5.

R78 · Membership. Plan; accessible from S67. Does the point (1.0;1) belong to 2x-y+3z=5?

R79 · Method. IPP; accessible from S66. In $\int x^2e^x\,\mathrm{d}x$, what factor drifts to simplify ?

R80 · Guidance. Rectangles; accessible from S66. For increasing f, the rectangles on the left Do they give a minor or a upper bound?

R81 · Case. Intersection; accessible from S70. After replacing a right in a plane, that means 0=1, and 0=0?

R82 · Error. Parameters; accessible from S70. Can the same parameter be imposed on two lines To search for their intersection?

R83 · Concentration. Chebychev; accessible from S73. With variance 4, which bound for $P(|X - E(X)| \geq 4)$?

R84 · Scope. NGL; accessible from S74. Does the law of large numbers guarantee a rapprochement Every rerun?

R85 · Angles. First; accessible from S4. Convert 90 degrees to radians and give the contact details of the associated point.

R86 · Circle. Signs; accessible from S6. On $[0; 2\pi]$, where is sin x positive or null?

R87 · Period. Trigonometry; accessible from S76. Give the smallest positive period of sin(3x).

R88 · Derivation. Composition; accessible from S76. Derivate cos(2x).

R89 · Equation. Circle; accessible from S78. Solve cos x=1/2 on $[-\pi; \pi]$.

R90 · Primitive. Trigonometry; accessible from S79. Calculate $\int_0^{\pi/2} \sin x \, \mathrm{d}x$.

Rituals proposed per meeting

Meeting	Questions	Operation
S2	R01, R02, R03	Answer without the course; explain an approach; Note the useful review.
S3	R04, R05, R06	Answer without the course; explain an approach; Note the useful review.
S4	R07, R08, R09	Answer without the course; explain an approach; Note the useful review.
S5	R10, R11, R12	Answer without the course; explain an approach; Note the useful review.
S6	R13, R14, R15	Answer without the course; explain an approach; Note the useful review.
S7	R16, R17, R18	Answer without the course; explain an approach; Note the useful review.
S8	R86, R85, R01	Answer without the course; explain an approach; Note the useful review.
S10	R02, R03, R04	Answer without the course; explain an approach; Note the useful review.
S11	R19, R20, R05	Answer without the course; explain an approach; Note the useful review.
S12	R21, R06, R07	Answer without the course; explain an approach; Note the useful review.
S13	R08, R09, R10	Answer without the course; explain an approach; Note the useful review.
S14	R22, R23, R11	Answer without the course; explain an approach; Note the useful review.
S15	R12, R13, R14	Answer without the course; explain an approach; Note the useful review.

Meeting	Questions	Operation
S16	R24, R15, R16	Answer without the course; explain an approach; Note the useful review.
S17	R17, R18, R86	Answer without the course; explain an approach; Note the useful review.
S19	R25, R85, R19	Answer without the course; explain an approach; Note the useful review.
S20	R26, R27, R20	Answer without the course; explain an approach; Note the useful review.
S21	R28, R21, R22	Answer without the course; explain an approach; Note the useful review.
S22	R31, R23, R24	Answer without the course; explain an approach; Note the useful review.
S23	R29, R25, R26	Answer without the course; explain an approach; Note the useful review.
S24	R30, R27, R28	Answer without the course; explain an approach; Note the useful review.
S25	R32, R33, R31	Answer without the course; explain an approach; Note the useful review.
S26	R34, R29, R30	Answer without the course; explain an approach; Note the useful review.
S27	R32, R33, R34	Answer without the course; explain an approach; Note the useful review.
S29	R35, R36, R37	Answer without the course; explain an approach; Note the useful review.
S30	R43, R38, R39	Answer without the course; explain an approach; Note the useful review.
S31	R40, R35, R36	Answer without the course; explain an approach; Note the useful review.
S32	R41, R42, R37	Answer without the course; explain an approach; Note the useful review.
S33	R43, R38, R39	Answer without the course; explain an approach; Note the useful review.
S35	R44, R45, R40	Answer without the course; explain an approach; Note the useful review.
S36	R46, R47, R48	Answer without the course; explain an approach; Note the useful review.
S37	R41, R42, R44	Answer without the course; explain an approach; Note the useful review.
S38	R49, R50, R45	Answer without the course; explain an approach; Note the useful review.
S39	R51, R46, R47	Answer without the course; explain an approach; Note the useful review.
S40	R48, R49, R50	Answer without the course; explain an approach; Note the useful review.
S41	R52, R53, R51	Answer without the course; explain an approach; Note the useful review.
S42	R52, R53, R01	Answer without the course; explain an approach; Note the useful review.
S43	R04, R02, R03	Answer without the course; explain an approach; Note the useful review.
S44	R54, R55, R56	Answer without the course; explain an approach; Note the useful review.
S47	R57, R54, R55	Answer without the course; explain an approach; Note the useful review.
S46	R56, R57, R05	Answer without the course; explain an approach; Note the useful review.

Meeting	Questions	Operation
S48	R60, R61, R07	Answer without the course; explain an approach; Note the useful review.
S49	R58, R60, R61	Answer without the course; explain an approach; Note the useful review.
S50	R59, R58, R06	Answer without the course; explain an approach; Note the useful review.
S51	R62, R59, R10	Answer without the course; explain an approach; Note the useful review.
S52	R64, R62, R08	Answer without the course; explain an approach; Note the useful review.
S53	R65, R64, R09	Answer without the course; explain an approach; Note the useful review.
S54	R63, R65, R11	Answer without the course; explain an approach; Note the useful review.
S55	R66, R67, R68	Answer without the course; explain an approach; Note the useful review.
S56	R63, R66, R67	Answer without the course; explain an approach; Note the useful review.
S60	R68, R13, R14	Answer without the course; explain an approach; Note the useful review.
S58	R12, R16, R15	Answer without the course; explain an approach; Note the useful review.
S59	R69, R17, R18	Answer without the course; explain an approach; Note the useful review.
S61	R70, R69, R86	Answer without the course; explain an approach; Note the useful review.
S62	R73, R70, R19	Answer without the course; explain an approach; Note the useful review.
S63	R71, R72, R73	Answer without the course; explain an approach; Note the useful review.
S64	R74, R75, R71	Answer without the course; explain an approach; Note the useful review.
S65	R76, R72, R74	Answer without the course; explain an approach; Note the useful review.
S66	R79, R80, R75	Answer without the course; explain an approach; Note the useful review.
S67	R77, R78, R76	Answer without the course; explain an approach; Note the useful review.
S70	R81, R82, R79	Answer without the course; explain an approach; Note the useful review.
S69	R80, R77, R78	Answer without the course; explain an approach; Note the useful review.
S72	R81, R82, R85	Answer without the course; explain an approach; Note the useful review.
S73	R83, R20, R22	Answer without the course; explain an approach; Note the useful review.
S74	R84, R83, R21	Answer without the course; explain an approach; Note the useful review.
S75	R84, R24, R23	Answer without the course; explain an approach; Note the useful review.
S76	R87, R88, R26	Answer without the course; explain an approach; Note the useful review.
S77	R87, R88, R25	Answer without the course; explain an approach; Note the useful review.
S78	R89, R28, R27	Answer without the course; explain an approach; Note the useful review.

Meeting	Questions	Operation
S79	R90, R89, R31	Answer without the course; explain an approach; Note the useful review.
S80	R90, R30, R29	Answer without the course; explain an approach; Note the useful review.
S81	R34, R32, R33	Answer without the course; explain an approach; Note the useful review.
S83	R35, R36, R37	Answer without the course; explain an approach; Note the useful review.
S85	R43, R38, R39	Answer without the course; explain an approach; Note the useful review.
S86	R40, R44, R41	Answer without the course; explain an approach; Note the useful review.
S88	R42, R45, R46	Answer without the course; explain an approach; Note the useful review.
S89	R47, R49, R50	Answer without the course; explain an approach; Note the useful review.
S91	R48, R51, R52	Answer without the course; explain an approach; Note the useful review.
S92	R53, R57, R56	Answer without the course; explain an approach; Note the useful review.

Corrections to fluency

R01. 10 and 13. The row is replaced by n.

R02. 1/8 and 4 by the rules on powers.

R03. No: 0.8 $\times 1{,}2 = 0.96$, an overall decrease of 4 per cent.

R04. The null vector, by Chasles.

R05. $(3;-3)$; midpoint $(5/2;1/2)$.

R06. The second is $-$Two times the first: they are colinear, of opposite meaning.

R07. 2,3,4. The final bound is excluded.

R08. Six: 8 $-3+1$.

R09. 31 ; direct addition or 2 $^{5}-1$.

R10. There is a rank n such as $u_n < 0$.

R11. Initialization at a specified starting level.

R12. No: x=-2 checks $x^2 = 4$ without merit 2.

R13. $2\vec{\imath} - 3\vec{\jmath} - \vec{k}$.

R14. No: her combinations remain in a vector plane.

R15. No: projection can create an apparent intersection.

R16. $-5 < -2x+3 \leq 1$; multiplication by -2 reverses inequalities.

R17. n+1, not n.

R18. $u_n = -1/(n+1)$.

R19. 6, because $6 = 6/2 + 3$.

R20. $1/((n+1)(n+2))$, strictly positive.

R21. No; it takes two non-colonial directions and one point.

R22. $2{,}9 < u_n < 3{,}1$.

R23. 10; in row 9, the difference is exactly 1/10.

R24. Divide up and down by n^2 ; The limit is 3.

R25. Both tend to 0 because the absolute value of the common difference / ratio is less than 1.

R26. No: $(-1)^n$ is bounded but its terms alternate between -1 and 1.

R27. No; it converges towards a limit not exceeding 5, e.g. a constant sequence equal to 2.

R28. $-x$, since $\sqrt{x^2} = |x|$.

R29. Negative; limit $-\infty$.

R30. $\sqrt{x+1} - 1 = x/(\sqrt{x+1}+1)$; the quotient tends to 1/2.

R31. Six, by the multiplicative principle.

R32. $3^3 = 27$; positions are distinguished.

R33. $3! = 6$; three choices, then two, then one.

R34. 0, by comparative growth; $\infty \times$ Only 0 does not conclude.

R35. $(x+2)e^x$.

R36. Factor 2 is missing: derivative 6 $(2x+1)^2$.

R37. 2 ; vertical variation 4 divided by horizontal variation 2.

R38. f is concave on I; this does not determine the sign of f'.

R39. No: $f(x) = x^4$ is convex on both sides of 0.

R40. 2, because g(x)=x+1 out of 1.

R41. Only in the range considered.

R42. $2/8 = 1/4$.

R43. $\binom{5}{2} = 10$; each ordered pair would have been counted twice.

R44. $\binom{6}{3} = 20$, by Pascal.

R45. The contrary event of "at least one" is "none"; cases are disjunct and exhaustive.

R46. 0,12 product of the industry.

R47. No; reference populations differ.

R48. At least two hits.

R49. Fixed number of tests, independence, two issues, same probability of success.

R50. $(1-p)^5$.

R51. 6 and 4.2.

R52. $1-P(X\leq 4)$ for X integer.

R53. $P(X\leq 5)-P(X\leq 1)$.

R54. $]-\infty;3[$.

R55. $\ln 4 = 2\ln 2$.

R56. Left limb is ln 2, right 0; false identity.

R57. $2x/(1+x^2)$ on $\mathbb{R}$.

R58. 0 ; place t=-ln x to obtain $-te^{-t}$.

R59. $n > \ln c/\ln q$; the meaning is inverse because ln q<0.

R60. 1.

R61. An obtuse or flat angle, in $]\pi/2;\pi]$.

R62. Two; check also that the candidate is non-zero.

R63. Belongs to the plane and direction joining the point to the orthogonal foot to the plane.

R64. $\frac{3}{2}e^{2x}$; the bypass provides control.

R65. No: the derivative of their difference is zero over the interval.

R66. $y(t)=3e^{2t}$.

R67. 90 minutes and 6 seconds.

R68. $1-0{,}1\times 1 = 0{,}9$ à t=0,1.

R69. No; the positivity of the integral applied to -f gives $\int_a^b f \leq 0$.

R70. 1, by the antiderivative x^2.

R71. Liters.

R72. 4; divide by length 3.

R73. 20 ; the factor is 2 ^{2}, translation does not change variance.

R74. $9/\mathrm{n}$.

R75. 2 because the standard deviation is proportional to $1/\sqrt{n}$.

R76. (1;2;0) and (1;-1;3).

R77. (2;-1;3), or any non-zero multiple.

R78. Yes, because 2+0+3=5.

R79. The polynomial x^2 ; its degree decreases.

R80. A minor, since each rectangle remains under the curve.

R81. No intersection; then right fully included.

R82. No, each right is driven independently.

R83. $1/4$.

R84. No; it concerns the probability of a fixed deviation when the size increases.

R85. $\pi/2$; (0;1).

R86. On $[0;\pi]$ and in point 2π.

R87. $2\pi/3$.

R88. $-2\sin(2x)$.

R89. $x = -\pi/3$ or $x = \pi/3$.

R90. 1, by the antiderivative -cos x.